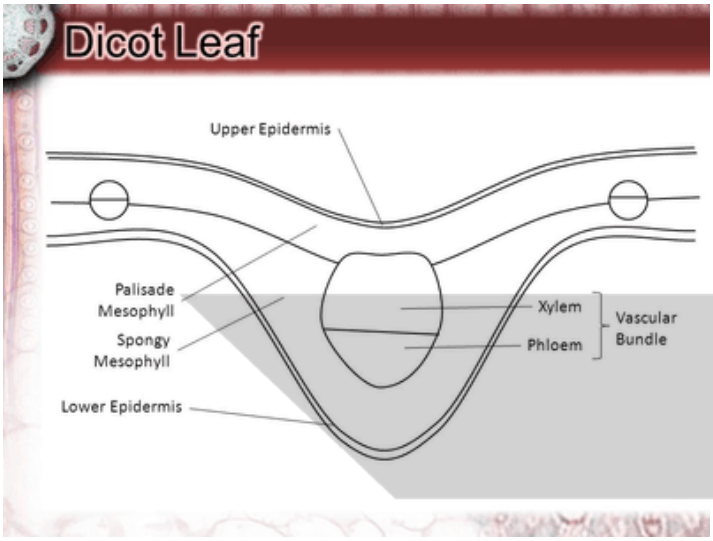
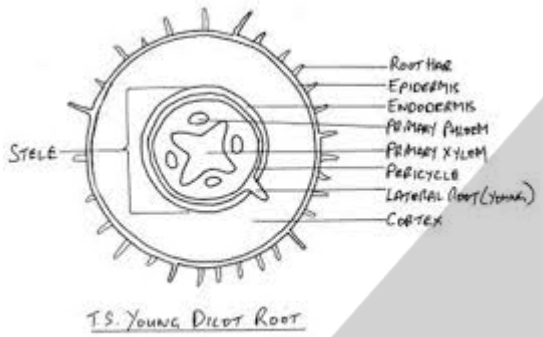
1. Transport in Plants

1.1. Structure of Transport Tissues

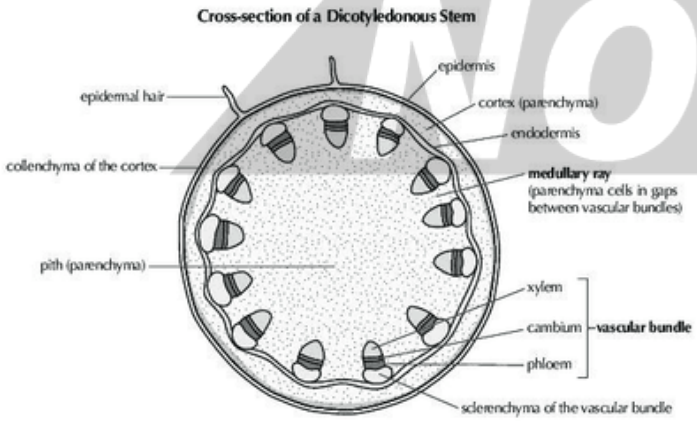
Plan diagram of leaf:



Plan diagram of root:



Plan diagram of stem:



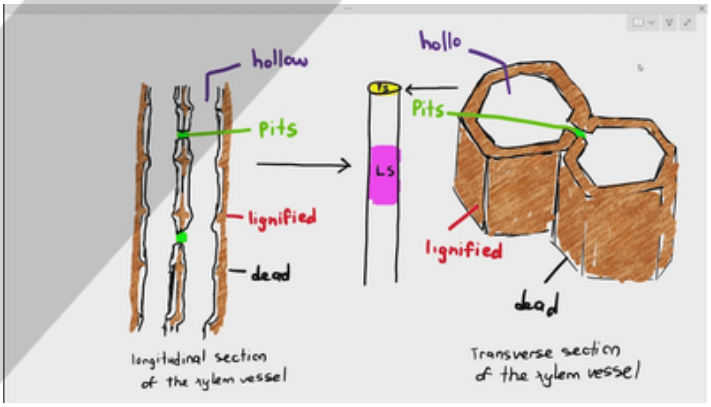
Arrangement of xylem and phloem (vascular bundle) in:

1. Leaf - Located in the middle, with the xylem being above the phloem
2. Root - Located in the centre, enclosed by the endodermis
3. Stem - Located on the periphery, with xylem being inwards and phloem being outwards

1.2. Xylem and Phloem

Xylem vessels:

- These are dead cells
- Form a long, narrow and hollow tube to increase capillarity
- They transport water unidirectionally (from roots to leaves)
- **Hollow / no cell contents and no end walls:** Little resistance to flow of water
- **Wide lumen:** Large amounts of water can be transported
- **Lignified cell walls:** Prevent collapse of vessels - mechanical support, impermeable to water
- **Cellulose cell wall:** Allows adhesion of water molecules to xylem walls as walls are hydrophilic
- **Pits:** Allow lateral movement of water and connect to all parts of plant
- **Narrow diameter:** For adhesion and to prevent air locks



Phloem:

- Contains 2 types of cells
  - **Sieve tube element:** It does not contain many organelles (no vacuole and nucleus) to decrease resistance to flow of phloem sap
  - **Companion cell:** Contains higher numbers of mitochondria and ribosomes as they are metabolically active cells
- Adaptations:
  - **Sieve pores:** Allow easy flow from one sieve tube element to the next
  - **Sieve plates:** Prevent sieve tube from bursting
  - **Little cell contents:** Little resistance to flow of sap
  - **Plasmodesmata:** Allows flow to/from companion cells for loading and unloading of sucrose
  - **Thin walls:** Rapid entry of water at source to build up hydrostatic pressure

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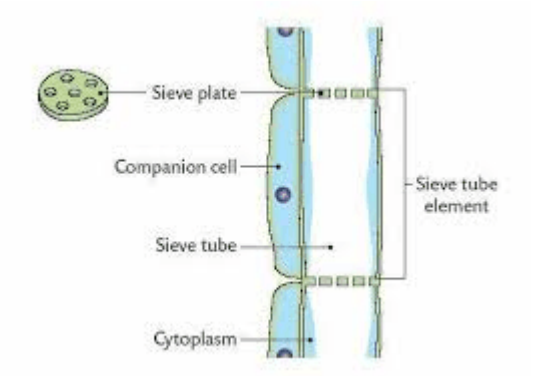
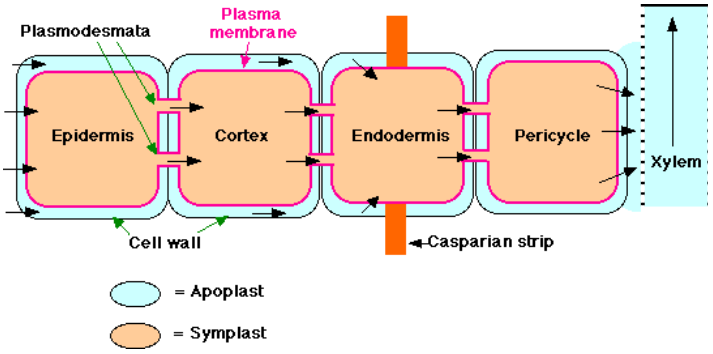


1.3. Transport Mechanisms

- Some mineral ions and organic compounds can be transported within plants dissolved in water
- Water reaches the root hair from the soil by:
  - The root hairs from the soil take up water
  - The soil has a relatively high-water potential, while the cytoplasm of the root hair cell has a relatively low water potential, as it contains many more inorganic and organic substances
  - Therefore, water moves into the root hair cells by osmosis
  - The large number of root hairs increases the surface area, allowing more water and ions to be taken up by the cells
  - Ions may be taken up with water by osmosis or separately through facilitated diffusion or active transport

Water reaches the xylem from the root hair through two main pathways:

- Apoplast pathway:
  - The movement of water through cell walls and intercellular spaces/ dead material by mass flow
  - Water can pass via apoplast pathway until it reaches the Casparian strip, as it is impermeable to water due to a suberin deposition in the cell wall
  - Water moves through the xylem takes the apoplastic route as xylem vessels are dead and contain no cytoplasm
- Symplast pathway:
  - This pathway involves the movement of water through the cytoplasm/ living material
  - Firstly, water passes through the partially permeable membrane
  - Then, the water passes through cytoplasm and vacuole before moving from one cell to another through the plasmodesmata



1.4. Transport in the Xylem

Transpiration:

- A process involving the loss of water in the form of water vapour from the aerial parts of the plant
- It is an inevitable consequence of gaseous exchange and thus photosynthesis, as stomata open, and this allows water vapour to diffuse
- Water vapour in air spaces diffuses out of the leaf through stomata down the water potential gradient
- Water evaporates from the cell walls of spongy mesophyll cells into the intercellular air spaces
- This causes water to move from the cell's cytoplasm into the cell wall
- Water from neighbouring xylem vessels (through pits) moves into mesophyll cells by osmosis, down a water potential gradient
- A cohesion-tension and transpiration pull is created at the top of the plant due to the evaporation of water, as the hydrostatic pressure at the top of the xylem is reduced
- This causes water to move up the xylem from roots to leaves
- Factors affecting the rate of transpiration:
  - The surface area of the leaf
  - Humidity
  - Wind speed
  - Number of stomata
  - Temperature
  - Light intensity
  - Thickness of cuticle
  - Water potential gradient between leaf and surrounding air

Hydrogen bonding in water causes:

- Cohesion:
  - Hydrogen bonds are formed between individual water molecules
  - Therefore, as one water molecule moves up the xylem, it pulls the other molecule along with it
  - This allows for water molecules to move up as a continuous stream

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- Adhesion:
  - Hydrogen bonds are temporarily formed between a water molecule and cellulose cell walls of the xylem.
  - This allows water molecules to continue moving upwards against gravity.

1.5. Transport in the Phloem

Translocation:

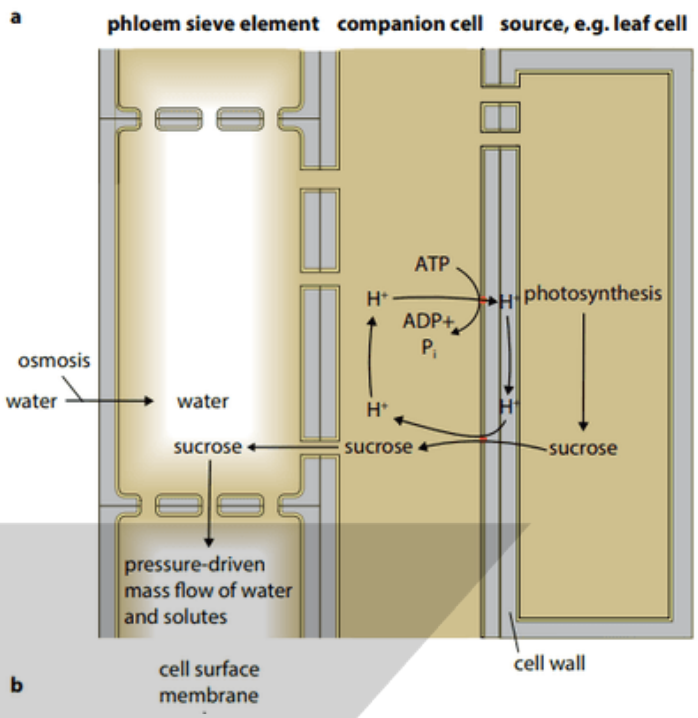
- The transport of soluble organic substances within a plant
- These are called assimilates, and they include sucrose and amino acids
- Sugars are transported as sucrose instead of glucose as glucose interferes with the water potential of cells

Loading of sucrose:

- Sucrose is loaded into the phloem tubes by companion cells at the source (leaves)
- This is made possible due to the transport proteins present in the cell surface membrane of companion cells
- The two transport proteins are:
  - The proton pump, driven by ATP, pumps H^+ ions (protons) out of the companion cell's cytoplasm into the cell wall (apoplast pathway), creating a high concentration of H+ ions in the cell wall
  - The H^+ ion-sucrose co-transporter then drives the movement of H^+ ions from a region of high concentration (the cell wall of companion cells) to a region of low concentration (cytoplasm of companion cell) via facilitated diffusion, along with sucrose, against its concentration gradient via secondary active transport/co-transport
- After sucrose is in the cytoplasm of the companion cell, it diffuses into the phloem sieve tubes through plasmodesmata, down a concentration gradient.
- When sucrose enters phloem sieve tubes, the water potential of the cells decreases due to an increase in solute.
- This causes water to move in from xylem vessels, causing an increase in hydrostatic pressure.

Mass flow: Movement of a liquid from high pressure to low pressure from source to sink PHLOEM

- Comparing High hydrostatic pressure SOURCE Low hydrostatic pressure. So creates a hydrostatic pressure gradient between SOURCE and SINK Thus creating a mass flow of sap solution towards the sink down the pressure gradient
- Unloading occurs at the sink (roots, tubers), where sucrose is used for metabolism or storage.
- Assimilates are transported in large quantities by mass flow and from a region of high hydrostatic pressure to a region of low hydrostatic pressure from the source to the sink.



1.6. Adaptations of Xerophytes

- Rolled leaf:
  - Increases humidity around stomata, reducing the water potential gradient
- Thick waxy cuticle:
  - Increases distance for diffusion, acting as a barrier for transpiration
  - The shiny surface reflects heat, lowering the temperature
- Hairs/trichomes on the surface:
  - Trap moisture to reduce water potential gradient
- Sunken stomata/ stomata in pits:
  - Moist air trapped in pits reduces water loss.
- No stomata on the upper surface:
  - Not exposed to sunlight, reducing the evaporation rate
- Small leaves, reduced to spines:
  - Reduce the surface area for transpiration

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